homework06

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Creating a Function

> Q.6 How would you generalize the original code above to work with any set of input protein structures?

Before creating a function code:

```
library(bio3d)
s1 <- read.pdb("4AKE") # kinase with drug
```

```
  Note: Accessing on-line PDB file
```

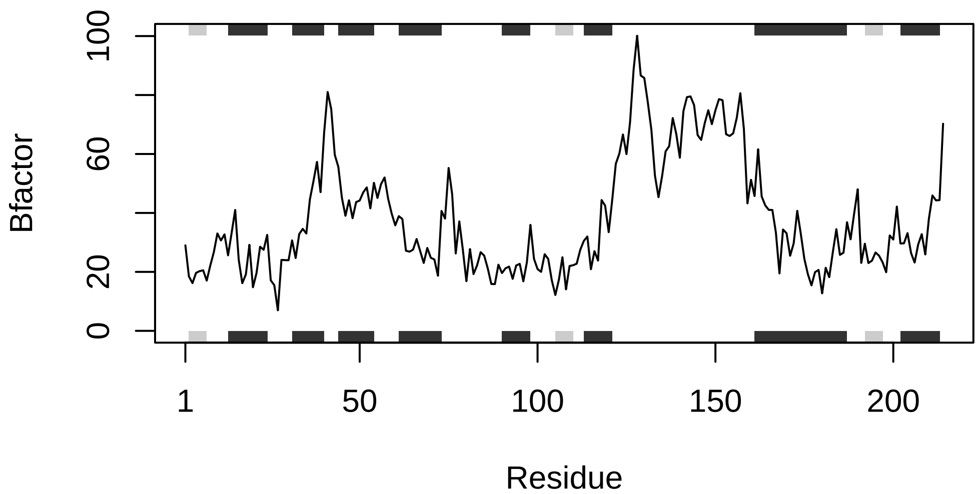
```
s2 <- read.pdb("1AKE") # kinase no drug
```

```
  Note: Accessing on-line PDB file
   PDB has ALT records, taking A only, rm.alt=TRUE
```

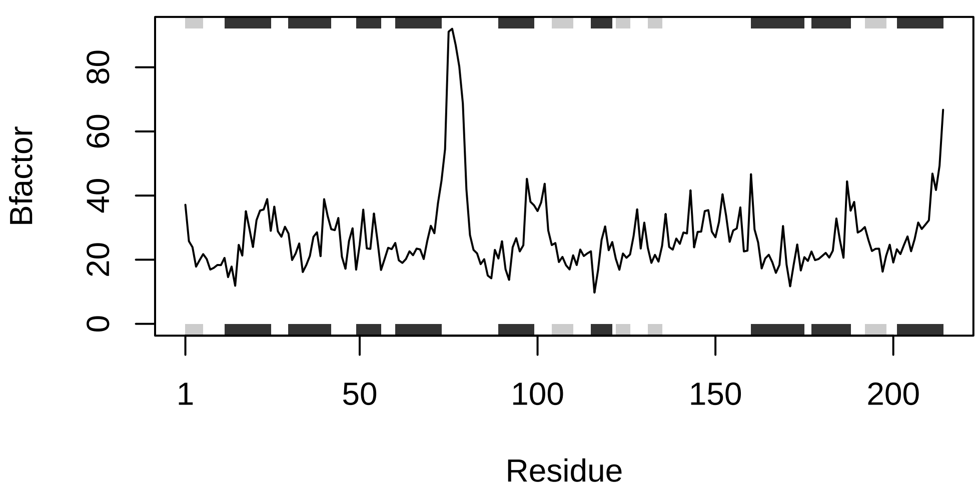
```
s3 <- read.pdb("1E4Y") # kinase with drug
```

```
  Note: Accessing on-line PDB file
```

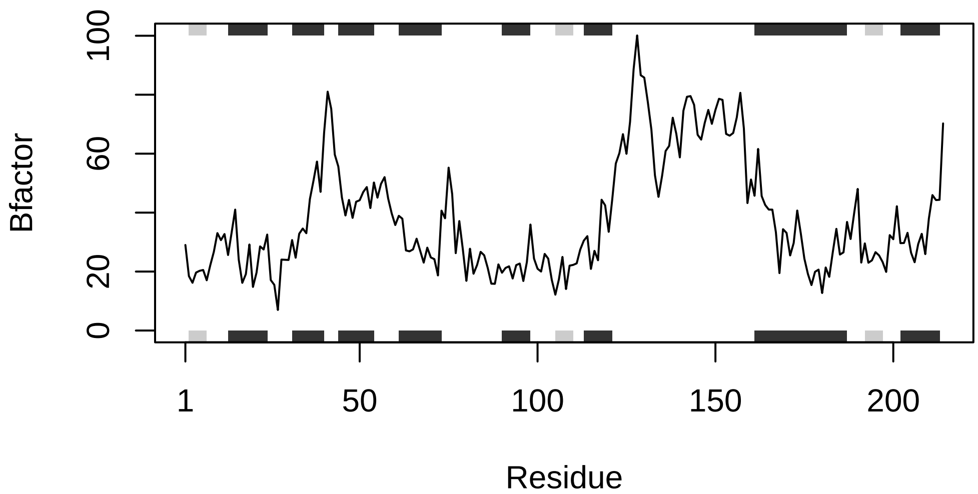
```
s1.chainA <- trim.pdb(s1, chain="A", elety="CA")
s2.chainA <- trim.pdb(s2, chain="A", elety="CA")
s3.chainA <- trim.pdb(s1, chain="A", elety="CA")
s1.b <- s1.chainA$atom$b
s2.b <- s2.chainA$atom$b
s3.b <- s3.chainA$atom$b
plotb3(s1.b, sse=s1.chainA, typ="l", ylab="Bfactor")
```




```r
plotb3(s2.b, sse=s2.chainA, typ="l", ylab="Bfactor")
```



```r
plotb3(s3.b, sse=s3.chainA, typ="l", ylab="Bfactor")
```



Process of creating function code:

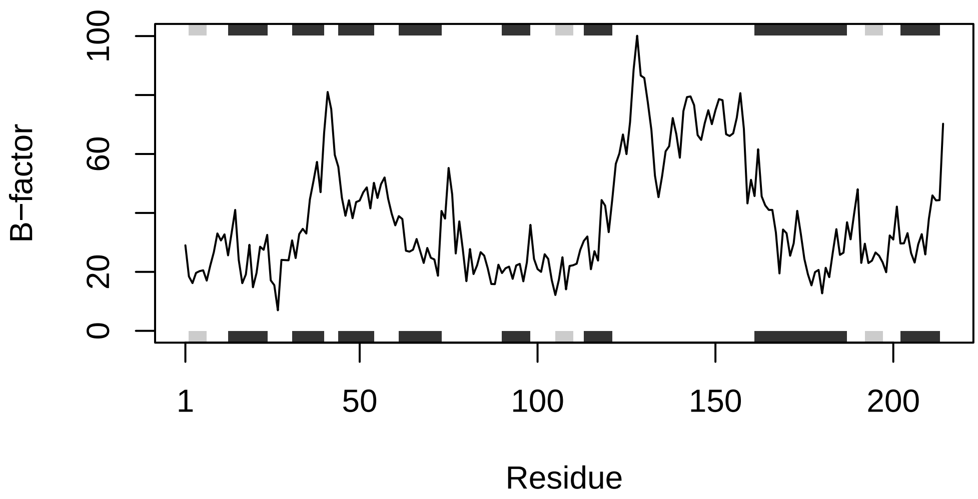
```r
library(bio3d)
analyze_protein <- function(pdb_id) {
  pdb <- read.pdb(pdb_id)
  pdb_chainA <- trim.pdb(pdb, chain="A", elety="CA")
  bvals <- pdb_chainA$atom$b
  plotb3(bvals, sse=pdb_chainA, typ="l", ylab="B-factor")
}
```

Testing it out:

```r
analyze_protein("4AKE")
```

```
  Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/qq/3hwqh6kd1gg5b7kq44_sjx600000gn/T//RtmpklUiqe/4AKE.pdb exists.
Skipping download
```

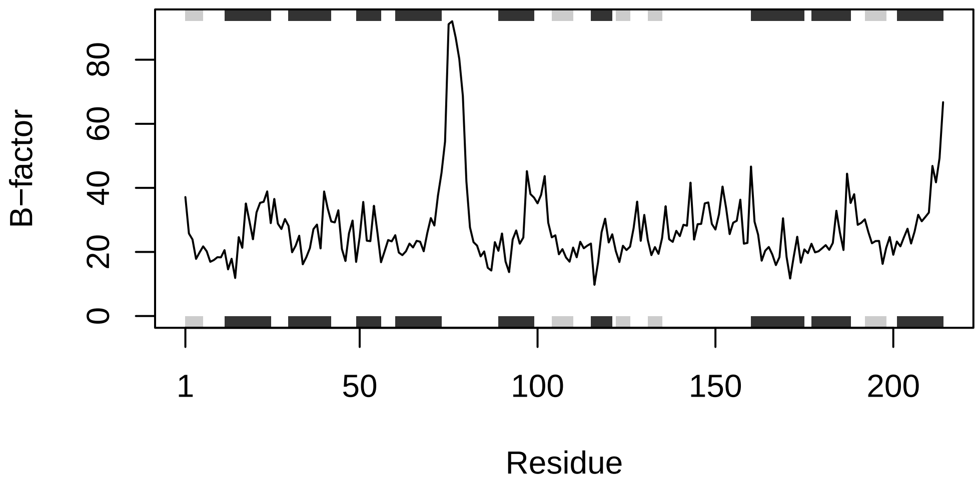



```
analyze_protein("1AKE")
```

```
  Note: Accessing on-line PDB file
```

```
Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/qq/3hwqh6kd1gg5b7kq44_sjx600000gn/T//RtmpklUiqe/1AKE.pdb exists.
Skipping download
```

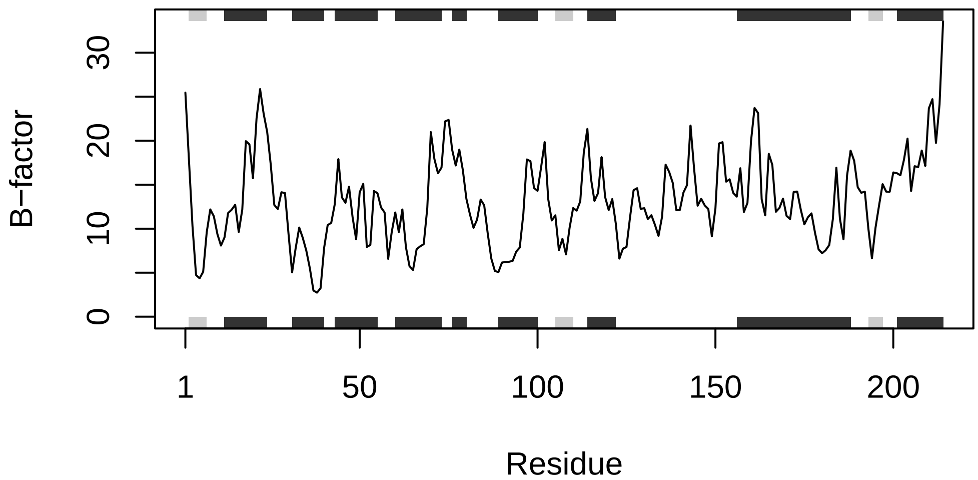
```
   PDB has ALT records, taking A only, rm.alt=TRUE
```




```
analyze_protein("1E4Y")
```

```
  Note: Accessing on-line PDB file

Warning in get.pdb(file, path = tempdir(), verbose = FALSE):
/var/folders/qq/3hwqh6kd1gg5b7kq44_sjx600000gn/T//RtmpklUiqe/1E4Y.pdb exists.
Skipping download
```



Inputs

1. `read.pdb(pbd.id)` is for reading the structure from the protein accession code within the PDB.
2. `trim.pdb()` is for trimming the structure to the chain and atom type. The default chain read is "A" and the default atom type to keep is "CA"
3. `pdb_chainA$atom$b` extracts the B-factors from the trimmed structure
4. `plotb3()` makes the plot for the B-factor. The `sse=` part puts the secondary structures like in the original code before the function.

Outputs

1. Primary output is the B-factor line plot for the structure inputed